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MILITARY SPECIFICATION

DISTRIBUTORS, WATER, TANK TYPE; TRUCK MOUNTED, FRONT AND REAR SPRAYBARS, COMMERCIAL

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

* 1. SCOPE

1.1 Scope. This specification covers tank type water distributors mounted on diesel-engine-driven, four-wheel, two-rear-wheel-drive, and six-wheel, four-rear-wheel-drive, and six-wheel-drive, commercial cab and chassis.

* 1.2 Classification. The water distributors shall be of the following types, as specified (see 6.2):

Type I - 1,000 gallon tank capacity, 16,000 pound (lb) gross vehicle weight rating (gvwr), 4 by 2.
Type II - 2,000 gallon tank capacity, 28,000 lb gvwr, 4 by 2.
Type III - 2,000 gallon tank capacity, 34,500 lb gvwr, 6 by 4.
Type IV - 2,000 gallon tank capacity, 45,000 lb gvwr, 6 by 6.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3825

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SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage, Vehicular, Ignition, Lighting, and Starting.
ZZ-H-451 - Hose, Fire, Woven-Jacketed, Rubber- or Cambric-Lined With Couplings.
ZZ-H-561 - Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water.

MILITARY

MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-D-771 - Distributors, Bitumen and Water, Packaging of.
MIL-N-12314 - Nozzles, Fire Hose, Water: Spray and Stream.
MIL-V-16721 - Valves, Foot, Low Pressure (For Water).
MIL-C-52404 - Connections, Hose, Fire and Water.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Automotive Vehicles.
FED-STD-595 - Colors.

MILITARY

MIL-STD-1223 - Administrative Wheeled Vehicles Treatment, Painting, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.
MS39153 - Coupling, Quick Disconnect, 3 Inch, For Construction Equipment.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

Interstate Motor Carrier Noise Emission Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

CONTRACTORS PUMP BUREAU (CPB)

Contractors Pump Standards.

(Application for copies should be addressed to the Contractors Pump Bureau, 1957 E Street, N.W., Washington, DC 20006.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The distributor shall consist of a water tank, water pressure system, front and rear header pipes with spray heads at each end, and the necessary controls, piping, and safety devices. The distributor shall be mounted on a commercial truck chassis as specified herein (see 1.2).

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3.2 First article. When specified (see 6.2), the contractor shall furnish one complete water distributor for first article inspection and approval (see 4.2.1 and 6.3).

* 3.3 Standard commercial product. The equipment shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the equipment being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3.1 Treatment, painting, and marking. As specified (see 6.2), by the procuring activity for the appropriate service, treatment, painting, identification marking, data plates, and warrant notice shall be in accordance with MIL-STD-1223. Color of finish coat shall be green No. 14064 of FED-STD-595.

3.3.2 Instruction plates. The distributor shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

* 3.3.3 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

3.3.4 Heavy-duty cooling system. A heavy-duty cooling system shall be furnished that shall maintain distributor chassis engine coolant at a temperature below the boiling point with distributor loaded to rated gvw, and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degree Fahrenheit (°F).

* 3.3.5 Radio interference supression. When specified (see 6.2), the distributor truck chassis shall be supressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be supressed to the same level as the vehicle chassis.

3.3.6 Towing devices. Not less than two hooks or loops for towing the distributor shall be furnished on the front of the distributor. Mounting shall include reinforcement to transfer loads directly to chassis rails.

3.3.7 Splash guards and shields. Rear wheels shall have metal splash shields ahead of the wheels and splash guards to the rear. Splash and stone throw protection shall be in accordance with SAE J682.

3.3.8 Engine hour meter. An engine hour meter, having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the distributor chassis engine to register accurately the number of hours of operating time.

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The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel.

3.3.9 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

* 3.4 Safety. All rotating or reciprocating parts and all parts subject to high operational temperatures, that are of such a nature or are so located as to be or become a hazard to the safety of the operating personnel, shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard.

3.4.1 Compliance. The vehicle and all accessories shall comply with all Federal Motor Carrier Safety Regulations and/or Federal Motor Vehicle Safety Standards and regulations in effect on the date of manufacture.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Weights, ratings, and dimensions.

3.7.1 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories, equipment, body, full complement of fuel, lubricants, and coolant.

* 3.7.2 Weight rating. The gvw rating shall consist of curb weight, operator (weight computed at 175 lb), and payload, and shall be not less than 16,000 lb for type I, 28,000 lb for type II, 34,500 lb for type III, and 45,000 lb for type IV.

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3.7.3 Ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification.

3.7.4 Cab-to-axle dimension. The chassis shall have a cab-to-axle dimension as recommended by the body manufacturer.

3.7.5 Overall width. The overall width shall be not more than 96 inches.

* 3.7.6 Overall height. No portion of the distributor unit shall have an overall height from the ground level greater than 102 inches.

3.8 Performance.

3.8.1 Speed and gradeability. High and low speed requirements shall be met with distributor loaded to specified gvw.

3.8.1.1 High speed gradeability. Distributor shall ascend the following continuous grades at 50 miles per hour (mph):

Type I - 2.5 percent grade.
Type II - 1.0 percent grade.
Type III & IV - 0.5 percent grade.

Gradeability requirements shall be met with the main transmission in direct drive, and when a multispeed axle is furnished, with the axle in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 truck ability prediction procedure (see 6.4).

3.8.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of recommended governed speed, and shall provide a distributor speed of not more than 4.5 mph for type I, 5.0 mph for type II, and 3.5 mph for type III and IV.

* 3.8.1.3 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 55 mph (minus 1 percent). Conformance to geared speed specified shall be calculated in accordance with the following formula:

$$\text{Maximum geared speed} = \frac{\text{Governed speed revolutions per minute}}{\text{Total reduction X tire factor}}$$

* 3.8.2 Brake performance. The service and parking brake shall conform to the performance requirements of Federal Motor Carrier Safety Regulations, sections 393.41, 393.52, or when applicable, Federal Motor Vehicle Safety Standard No. 121, under all loading conditions up to the specified payload capacity.

3.9 Chassis components.

* 3.9.1 Engine. The engine furnished shall be the manufacturer's standard or optional diesel engine for the commercial model distributor chassis which meets or exceeds the requirements of this specification. Engine shall be liquid cooled, compression ignition, two or four stroke cycle diesel engine, with not less than six cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J816.

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3.9.1.1 Oil filter. Full flow type oil filter shall be furnished.

3.9.1.2 Governor. Engine governor shall be furnished and set and sealed to limit engine to engine manufacturer's maximum recommended operating speed.

3.9.1.3 Air pollution control. All distributors shall comply with EPA regulations governing control of air pollution from New Motor Vehicles and Engines applicable on the date of manufacture to nonmilitary, nonexport motor vehicle products. When specified (see 6.2), distributor shall additionally comply with the State of California regulations governing air pollution control applicable on the date of manufacture.

* 3.9.1.4 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulations, section 393.94. The vehicle exterior sound level shall conform to the noise emission standard of the EPA when tested in accordance with regulations of DoT, part 325. When applicable the exterior sound level shall conform to California Vehicle Code.

3.9.2 Electrical equipment.

3.9.2.1 Charging system. The distributor shall be equipped with a 12-volt (V) negative ground system. An alternator of not less than 60 ampere (A) rated capacity shall be provided, which shall be capable of maintaining an adequate supply of electrical power, under all operating conditions, for all 12V accessory components of the distributor operating simultaneously.

3.9.2.2 Lighting and wiring. The distributor lighting and wiring shall conform to and be installed in accordance with DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.19, 393.20, 393.22 through 393.26, 393.27 through 393.31, and 393.33.

* 3.9.2.3 Starting system. A 12 or 24V direct current starting system with 12V lighting system, and with no less than 60A alternator, shall be furnished. Engine starting equipment shall include an ether starting or glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. Measured shot type systems shall be manually operated from the driver's compartment, shall have a reservoir of not less than 12 fluid ounces, and shall be inoperative with the engine warm.

* 3.9.2.4 Batteries. Each battery shall be of 12V potential. The total reserve capacity shall be not less than 230 minutes and the total cold cranking rating at 0°F shall be not less than 1,200A for military services and 1,600A for civil agencies measured in accordance with SAE J537. Batteries furnished shall be dry charged in accordance with W-B-131, without electrolyte, with sealed caps to prevent the intrusion of atmospheric moisture.

3.9.3 Fuel system. Fuel system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.65.

* 3.9.3.1 Air cleaner. A dry type air cleaner with replacement element shall be furnished. The air cleaner shall provide the maximum air requirements for the engine furnished.

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* 3.9.3.2 Fuel tank. Fuel tank shall be manufacturer's standard for the distributor furnished. When more than one tank is furnished, means shall be provided to assure equalized fuel level in both tanks.

3.9.4 Exhaust system. The exhaust system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.83. All horizontal portions of the exhaust system, which project to the rear of the cab, shall be provided with a cover plate. Cover plate shall form a shield that will shed fluids, precluding spilled fuel from coming in contact with the exhaust system.

3.9.5 Automatic transmission. The vehicle shall be provided with an automatic transmission. The transmission furnished shall provide continuous drive. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios for type I, and not less than five forward gear ratios for types II, III and IV. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with SAE J704 power takeoff opening(s). The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

3.9.5.1 Transfer case. Unless the transfer case is equipped with devices which compensate for differential torque and speeds between front and rear axles, the transfer case shall provide for driver selection of either four-wheel or six-wheel drive. When furnished, inter-axle compensating devices shall provide for positive transfer of power to all driving axles. A one or two speed transfer case shall be furnished.

3.9.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.9.7 Frame. Chassis frame shall be manufacturer's standard for the distributor furnished.

3.9.8 Suspension. Chassis shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with distributor loaded to specified gvwr. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Chassis shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

3.9.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with distributor loaded to specified gvwr.

* 3.9.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with wide base type tires and wheels. Wheels shall be interchangeable without the use of an adaptor. Tires shall conform to TRA recommendations with a size designation system the same as the TRA. Rim and tire ratings shall conform to TRA recommendations for the type and size tires furnished. Tire and rim sizes shall be the same for all wheels on each vehicle. Centerline of front and rear wheels shall be equidistant, $\pm$1 inch.

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3.9.10.1 Tires. Tires shall be steel belted radial or, when specified (see 6.2), bias ply. Tires shall be tube or tubeless type, with nondirectional mud and snow tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at the ground, with the vehicle loaded to specified gvw.

3.9.10.2 Inner tubes. For tube tires, inner tubes shall be of heavy duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with TRA recommendations.

3.9.10.3 Spare tire assembly. A spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size and tread design and load range (ply rating) as the tires furnished on the vehicle.

3.9.10.4 Carrier for spare tire assembly. A carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the veicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion resistant material. The carrier design shall enable removal or mounting of spare wheel assembly using only tools furnished with the vehicle.

* 3.9.11 Tools. Each vehicle shall be furnished with tools required for exchanging mounted tire assembly with the spare assembly and shall include at least a hydraulic jack, jack handle, and wheel nut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.9.12 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40 through 393.42, 393.45 through 393.51, and as specified herein.

* 3.9.12.1 Service brakes. Type I distributor shall be equipped with vacuum assisted, hydraulic actuated, four wheel service brakes of the manufacturer's standard type. Type II, III, and IV distributors shall be equipped with full-air brakes. Type II, III, and IV braking system, complete with all necessary components, shall include:

a. Air compressor, unloader head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7 1/4 cubic feet per minute (cfm) for type III.
b. Air storage reservoir(s) with not less than 2,000 cubic inches total capacity for type II, and not less than 3,000 cubic inches for type III; each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank.
c. Foot control, suspended or treadle type.
d. Air control valves.
e. Service brake STOP lamp switch.
f. Air-pressure gage, visible to the driver.

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g. Low-air-pressure warning, visible and audible.
h. Alcohol aspirator with unbreakable transparent container.
i. Automatic moisture ejector.

3.9.12.2 Air brake emergency system. When specified for types II, III, and IV (see 6.2), an emergency brake system shall be furnished. The system shall be capable of being manually applied by the operator seated in the driver's seat. A manual or mechanical means to release the brakes shall be provided when compressed air is not available to release the brakes. The emergency system shall conform to the requirements of the California Vehicle Code, section 26058. The emergency system, when provided, shall also serve as a parking brake.

3.9.13 Cab. Manufacturer shall furnish a standard full-width cab. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with at least the curbside door equipped with external, key-operated lock. Cab shall have upholstered, fullwidth, adjustable seat and back, or individual, adjustable, driver's seat and individual passenger seat. Interior dome lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be furnished and installed.

3.9.14 Windshield wipers and washers. Distributor shall be equipped with dual windshield wipers and washers. Windshield wipers shall be variable or multispeed type and operated by either air or electric motor(s).

3.9.15 Steering. Manufacturer's standard power steering shall be furnished.

3.9.16 Heater and defroster. Hot water heater shall be provided. Heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to floor and to defroster louvers. Heater shall be complete with blower and mounted controls convenient to the driver.

3.9.17 Front bumpers and rear end protection. Distributor shall be equipped with channel type front bumper. Rear end of distributor shall be protected in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.86.

* 3.9.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver while seated in the operator's seat. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from drivers compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

3.9.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment snall be furnished:

a. Outside rearview mirrors, mounted on each side of the cab, each with not less than 50 square inches of reflective area for type I and 80 square inches for types II, III, and IV.

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b. Key-operated ignition switch.
c. Voltmeter or charging indicator.
d. Fuel gage.
e. Oil pressure gage or indicator.
f. Engine temperature gage or indicator.
g. Speedometer with recording odometer.
h. Dual sunvisors.
i. Front dooor mounted armrest on driver's side.
j. Driver's compartment ventilator other than window.
k. Manufacturer's standard electric or air horn.

3.10 Distributor components.

3.10.1 Tank. Tank shall be of elliptical cylinder shape and shall be fabricated from steel sheets. Steel sheets used in the tank cylinder shall be of not less than 0.1345-inch (U. S. revised standard gage No. 10) nominal thickness, and steel sheet used in the heads shall be not less than 0.1644-inch (U. S. revised standard gage No. 8) nominal thickness. The heads shall be flat and reinforced, concave, or convex, and all seams shall be welded and watertight. Tank shall have transverse, steel, surge plates made of not less than 0.1345-inch (U.S. revised standard gage No. 10) nominal thickness sheet steel. The spacing between surge plates and between surge plate and tank ends shall be not more than 48 inches. The transverse surge plates shall be welded in place. The surge plates shall be the full size of the tank cross section and shall be dished or reinforced to withstand the surge pressure. The surge plates shall permit free flow of water at the bottom, venting at the top, and access to all sections of the tank. Access holes in the surge plates shall be staggered. Tank shall be provided with a manhole not less than 18 inches in diameter. The manhole shall be located at the top of the tank and shall be equipped with a hinged, watertight, quick-clamp-type cover. A 3-inch unrestricted opening for overflow and for venting to the atmosphere at all times shall be provided and shall be protected against surge discharge. The 3-inch opening shall be located not more than 2 feet from the transverse centerline of the tank. Tank shall be provided with a sump to allow complete draining through the pump. The tank capacity shall be as specified.

3.10.2 Tank gage. A mechanical-type or sight-type gage shall be provided. The gage shall be calibrated in gallons from zero to tank capacity in increments of not more than 25 gallons. The gage shall have an accuracy so that the indicated amount of water shall be within 25 gallons of the actual amount of water in the tank. The gage shall be protected from breakage. The gage shall be located so as to be readable from inside the cab.

3.10.3 Distributor mounting. The distributor body shall be mounted with not less than three U-bolts or twin studs, on each side, having 5/8-inch minimum body diameter, or other equivalent method(s). Tie plates shall be at least 1/2 inch thick. Vehicle chassis frame shall be braced, using hardwood blocks or steel spacers, at each mounting point unless mounting point is located at a full depth, frame crossmember. Blocks shall incorporate a keeper retention. Two sheer bolts shall be provided, one on each side of rear portion of body subframe, to maintain body alinement on vehicle chassis. A

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hardwood breaker strip, of not less than 1 1/16-inches finished thickness, shall be installed between longitudinal main sills and vehicle chassis frame. Breaker strips shall have a taper of 1 inch in 18 inches at forward end. Forward body mounting bolts shall be located to the rear of the tapered portion of the breaker strips. When body is secured by brackets attached to chassis frame rails, attachments must be bolted to the web of the frame rails.

* 3.10.4 Header pipes and spray heads. The truck shall have a header pipe, with a four-way adjustable spray head at each end, mounted at the front and rear of the vehicle at bumper height. The spray heads shall be adjustable, without the use of special tools to produce a variable spraying range, from a minimum width not more than 8 feet to a maximum width of not less than 24 feet. Each spray head shall be controlled with a pneumatic or hydraulically operated control valve. Each valve shall be individually controlled from inside the truck cab at a location readily accessible to the driver. The valve controls shall be identified as to functinon and operating positions.

* 3.10.5 Pump unit. The pumping unit shall consist of a full torque power takeoff installed on the truck, a hydraulic fluid reservoir, a constant displacement hydraulic motor, a self-priming centrifugal pump, hydraulic lines, filters, safety devices, and controls. The pump unit shall be accessible for service and maintenance. The power takeoff controls shall be located in the truck cab. A red warning light, properly identified and located on the truck instrument panel, shall be provided to indicate power takeoff engagement.

3.10.5.1 Pump. The pump shall be an integral self-priming, 4-inch suction and discharge, single-stage, single-suction, centrifugal pump. The impeller and renewable wearing rings shall be bronze. The impeller shaft shall be stainless steel, and shall turn in antifriction bearings. The case shall be close grained cast iron, with ready means for removal of the impeller. Pump shall be provided with necessary vent and drain facilities. When operated against a total head of 30 feet and using a 25-foot static suction lift, the pump shall deliver not less than 350 gallons per minute (gpm). The pump shall meet the capacity requirements of the model 40-M pump as specified in the CPB Contractors Pump Standards.

3.10.5.2 Pump fittings and connections. The necessary pipe and fittings shall interconnect the pump to the distributor in such a manner that the following operations will be possible:

a. Pump water from an outside source, such as lakes, ponds, or streams, into the tank.
b. Pump water from the tank to the spray heads.
c. Pump water from the tank to a 1 1/2-inch hose outlet without water entering the header pipe.
d. Pump water from an outside source, without water entering the header pipe or tank, to a 1 1/2-inch hose outlet.

A removable, full-opening-type strainer of stainless steel shall be placed in the line ahead of the pump so as to filter all water from tank or outside source prior to entry into the pump. The strainer shall be removable without

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removing any fittings or connections other than the 4-inch inlet dust cover. Piping from the pump to the rear header pipe shall include a 1 1/2-inch hose outlet with external 1 1/2-inch 9 American National Hose coupling threads. The suction side of the pump shall be furnished with a 4-inch coupling conforming to MS39153 (faucet end) for connecting the suction hose. The 4-inch inlet and the 1 1/2-inch outlet shall be provided with a dust cover secured by a chain or wire rope.

3.10.6 Controls. Operating controls shall be located at a control station located on the left side (street side) of the vehicle and shall be readily accessible to the operator standing on the ground. Controls shall be clearly identified, for their intended operation and function, on permanently affixed plates. Controls shall permit the following operations:

a. Fill tank from an outside source using pump.
b. Pump water from tank to either or both front and rear header pipes.
c. Pump water from tank to a 1 1/2-inch diameter connection for hose attachment without water entering the header pipes.
d. Empty tank through spray heads by gravity.
e. Pump water from outside source, without water entering header pipes or tank, to a 1 1/2-inch hose outlet.

An operating instruction plate shall be provided describing step-by-step procedures required to perform all operations specified herein and shall include a flow and valve diagram. The plate shall be permanently affixed to the distributor so as to be discernible from the control station.

* 3.10.7 Application and distribution. The rate of application shall be at 120 gpm, and 290 gpm when distributed through the spray heads adjusted to produce the maximum width of spread as specified in 3.10.4. The distributor shall apply water through the header pipes and spray heads to the surface being sprayed at all combinations of specified widths and rates without streaking or puddling.

3.10.8 Pressure and vacuum gages. A pressure and vacuum gage shall be installed in the truck cab and at the control station. The pressure gage shall be connected to the discharge side of the pump to indicate pressure in the line, and shall be calibrated in a 5-lb increments from 0 to not less than 110 percent of the maximum working pressure. A vacuum gage, similar to the pressure gage, shall be installed on the suction side of the pump.

3.10.9 Suction and discharge hoses. A suction hose assembly and a discharge hose assembly shall be provided. Provisions shall be furnished to stow the hose assemblies.

3.10.9.1 Suction hose assembly. The suction hose assembly shall consist of three 10-foot sections of 4-inch hose conforming to ZZ-H-561, grade B, type I. Two sections of the hose shall be fitted with 4-inch couplings conforming to MS39153 (hose-end on one end, faucet-end on the other end). Each faucet-end shall be provided with a dust cover secure by chain. One double adapter shall be furnished. The third section of the hose shall be fitted on

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one end with a hose-end, quick-detachable coupling and on the other end with a 4-inch pipe nipple with American National tape pipe (thread). A 4-inch foot valve, conforming to MIL-V-16721, shall be provided for use with the suction hose.

3.10.9.2 Discharge hose assembly. The discharge hose shall be a 1 1/2-inch inside diameter, 25 feet long, and shall conform to ZZ-H-451. A nozzle, conforming to MIL-N-12314, type III, class A, 1 1/2-inch with 5/8-inch tip opening, shall be provided. A reducing connection conforming to MIL-C-52404, type XV, class A, size 2 1/2-inch NH by 1 1/2-inch NH, shall be provided to connect the nozzle end of the hose to a fire hydrant. Fixtures with hose threads shall be mounted in the toolbox to provide safe storage for the connector and the nozzle.

3.11 Toolbox. The vehicle shall be provided with a toolbox made of 0.0747-inch (U. S. revised standard gage No. 14) nominal thickness sheet steel. The toolbox shall have an overlapping hinged lid which prevents the entry of rain and dirt. The lid shall be held closed with two hooktype fasteners. The toolbox shall be of a size to provide stowage space for necessary tools and fixtures specified in 3.10.9.2.

3.12 Access ladder. A ladder shall be furnished to provide access from the ground to the manhole on top of the tank.

3.13 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Pressure grease fittings shall be of the hydraulic type with surface check ball and, when applied to sealed bearings, shall have pressure release to protect the seals.

3.14 Servicing and adjusting. Prior to acceptance of the truck mounted distributor by the Government, the contractor shall service and adjust the distributor for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

1. Inflation of all tires.
2. Adjustment of brakes.
3. Proper functioning of all lighting and electrical systems.
4. Wheel alinement.
5. Adjustment of engine to include tune-up.
6. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
7. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The distributor shall be conspicuously tagged to identify the lubricants and their temperature range.

* 3.15 Painting, tank.

a. Tank exterior and components. The first coat of rust inhibiting primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be wi' manufacturer's current materials according to manufacturer's

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current processes except that the total dry film thickness shall be not less than 2.5 mils. The paint shall be free from runs, sags, orange peel, or other defects. Color of the finish coat shall match color chip 14064, green gloss of FED-STD-595.

b. Tank interior surfaces. The tank interior shall be sand blasted and after cleaning shall be painted with two coats of rust inhibiting paint.

* 3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.16.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.17 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from comtaminants such as oil, grease, welding slag, and spatter, loose mill scale, water, dirt, corrosion products, or any other interfering substances. As soon as practible, after cleaning, and before any corrosion product or other coating interfering material can result, the surface shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as

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soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. Color of the finish coat shall conform to FED-STD-595. The end item, allied equipment, and attachments shall be the same color.

3.18 Lifting and tiedown attachments. Lifting attachments shall have at least three eyes with 3-inch inside diameter. The eyes shall be located to provide uninhibited lifting at a convergence not to exceed 24 feet above the lowest point of the vehicle. The lifting attachment and its components shall have a yeild strength not less than 5 times the proportionate empty vehicle weight imposed on each attachment. The entire vehicle shall remain in an approximately level position when being lifted. Tying down attachments shall be located symmetrical about the vehicle so as to restrain the spring weight on the chassis of the vehicle. At least two attachments may also serve as tying down attachments. Instruction plate(s) shall be furnished showing the lifting attachments and lifting slings, the capacity of each attachment and the required length and size of each sling cable. A silhouette of the vehicle showing the center of gravity shall be provided on the plate. Tying down attachments shall be identified by stenciling or other suitable marking, clearly indicating the intent of the attachments for the tiedown of the equipment when being shipped.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one distributor when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified, the preproduction pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard

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production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each distributor shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

* 4.4 Tests.

4.4.1 Weight. The distributor shall be weighed to determine curb weight and distribution of curb weight on front and rear axle(s). The imposed loading on front and rear axle(s) shall be computed using the curb weight, the operator weight at 175 lb, and the payload (see 3.7.2 and 3.7.3), to provide the specified gvwr. The calculated imposed loads on front and rear axle(s) will be utilized to ascertain that the suspension, axles, and tires are of adequate capacity to meet requirements of 3.9.9, 3.9.10, and 3.9.11.1.

4.4.2 Service brake test. Service brakes shall be tested to verify conformance to the requirements of 3.8.2.

4.4.3 Parking brake test. Parking brake shall be tested to verify conformance to the requirement of 3.8.2.

4.4.4 Road test. The distributor, loaded to provide not less than rated gvw, shall be driven at speeds up to 55 mph over improved roads (see 6.5), for not less than 30 miles, and at speeds of 20 mph over unimproved roads (see 6.5), for not less than 10 miles. Upon completion of the test, the distributor and equipment shall be examined for leaks, warping, distortion, or malfunction.

4.4.5 Test for radio interference supression. When radio interference supression is required, the distributor truck chassis shall be tested to determine conformance to 3.3.5. In lieu of tests to determine conformance to SAE J551, the manufacturer may furnish a certification that the distributor truck chassis meets the requirements, together with a list of the supression devices intalled. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

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4.4.6 Spray test. Operate the distributor to discharge water through the spray heads adjusted to maximum specified width (see 3.10.4), at the following variable rates:

a. With either the front or the rear header bar operating independently, the discharge rates shall be 120 and 290 gpm.
b. With both header bars operating, the discharge rates shall be 120 gpm and not less than 290 gpm at maximum pump capacity. Then operate the distributor to apply water over the minimum specified width at a rate of 120 gpm. Distribute water over a distance of not less than 100 feet at a speed of approximately 1 mile per hour at maximum width and rate and at minimum width and rate, and observe applied water after each application for streaking or puddling. During all portions of sprinkler test, observe the sprinkler system for leaks.

4.4.7 Pumping system operation. Perform the following operations to verify conformance to the requirements of 3.10.5.2 and 3.10.6:

a. Pump water from outside source into the distributor tank utilizing all three 10-foot sections of suction hose.
b. Pump water from distributor tank to each header pipe separately.
c. Pump water from outside source to 1 1/2-inch outlet and through discharge hose assembly at maximum rate without water entering header pipes or tank.
d. Drain distributor tank through spray heads by gravity.

Inability of the distributor to perform any of these operations in accordance with requirements specified herein shall constitute failure of this test.

4.4.8 Leak test. Distributor tank shall be tested for leaks by air or hydraulic pressure of not less than 3 pound-force per square inch gage. Air pressure, if used, shall be maintained for a period of at least 5 minutes during which the entire surface of all joints under pressure shall be coated with a solution of soap and water or other material suitable for detecting leaks. The piping shall be tested at the functional discharge pressure. Any leak shall be corrected and upon completion of repairs, shall be retested.

4.5 Production distributor test. Operate each distributor to demonstrate satisfactory operation of pump, valves, controls, spray heads, and gages and make necessary adjustments. Check tank and plumbing for leaks. Failure of the production unit to meet requirements of the contract shall be cause for the Government to refuse acceptance of all units under the contract until corrective action has been taken.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-D-771. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

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5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-D-771 with the level of preservation and the level of packing as specified (see 6.2). P-21 shall be substituted for P-3.

* 6. NOTES

6.1 Intended use. The water distributors are primarily intended for use in soil stabilization and soil compaction operations. The water distributors can also be used for transporting or transferring water, for application of water for dust alleviation, for sump pumping, for washing equipment, and as auxiliary fire fighting equipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of water distributor required (see 1.2).
c. When first article is required (see 3.2).
d. Identification of appropriate, service for treatment, painting, and marking (see 3.3.1).
e. When radio interference suppression requirements and test limits of SAE J551 are required (see 3.3.5).
f. When fungus resistant coating is required (see 3.3.9).
g. When distributor shall comply with the State of California pollution control regulations (see 3.9.1.3).
h. When bias ply tires shall be furnished (see 3.9.10.1).
i. When air brake emergency system is required for types II, III, and IV distributor (see 3.9.12.2).
j. When a preproduction pack inspection is required and time frame required for submission (see 4.2.1 and 4.6).
k. Level of preservation and level of packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.4 Performance prediction. Completed copies of SAE J688 form and computation for speed requirements for each model furnished under contract should be submitted as specified in the contract. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The SAE Work Sheet Item 1 should include distributor chassis model number, wheelbase, engine model, and transmission model number. The factors used in predicting distributor ability (see 3.8.1.1) are established as follows for the corresponding SAE J688 tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.	
Table 2	- Altitude Factor.	1.00
Table 3	- Rolling Factor.	1.613
Table 4	- Area Factor.	0.173
Table 5	- Velocity Factor.	250.0
Table 6	- Altitude Factor.	1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit gvw (to nearest, higher, 1,000 lb) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For gvw and engine speed beyond the range of this table, factors shall be extrapolated.	
Table 8	- Grade Factor.	0.75
Table 8A	- Correction Factor.	Not required
Table 9	- Road Factor.	0.0

6.5 Definitions.

6.5.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or asphalt paved highway.

6.5.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.7 Title change. The title of this specification was changed from "Distributors, Water, Tank Type; Truck Mounted, Sprinkler or Nozzle Spraybar, Commercial" to "Distributors, Water, Tank Type; Truck Mounted, Front and Rear Spraybars, Commercial".

Custodian:

Navy - YD

Preparing Activity:

Navy - YD

(Project 3825-N171)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side) 3825-N171

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-28531C(YD)	DISTRIBUTORS, WATER, TANK TYPE; TRUCK MOUNTED, FRONT AND REAR SPRAYBARS, COMMERCIAL

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS (Street, City, State, ZIP Code)

4. TYPE OF ORGANIZATION (Mark one)

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional

b. WORK TELEPHONE NUMBER (Include Area Code) – Optional

c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional

8. DATE OF SUBMISSION (YYMMDD)

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

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